

Q. P. Code: 23079**2 ½ Hours****Total Marks: 75**

1. Attempt **all** questions.
2. **All** questions carry **equal** marks.
3. Draw **neat labelled diagrams** wherever necessary.
4. Use of **log tables** and **non-programmable calculator** is **allowed**.
5. For **Q 2, Q 3 and Q 4** attempt A and B **OR** C and D.

Q 1 Do as directed (Any fifteen)**15**

1. Give an example of a granulocyte.
2. Name the dendritic cell specific to skin.
3. What is the site for T cell maturation?
4. Give the full form of MALT.
5. Fill in the blank: CD 8⁺ cells are also called as _____.
6. Give an example of a cell that contains vesicles with histamine.
7. State true or false: C4b2a is also known as C5 convertase.
8. State true or false: Pro- T cells express CD 7 on their cell surface.
9. Fill in the blanks: Peptide binding cleft of MHC class I is made of ____ and ____ domains.
10. State true or false: MHC class II is seen on the surface of all nucleated cells.
11. State true or false: TCR bind antigens in association with MHC.
12. State true or false: T_H cells secrete cytokines to activate B cells.
13. Define: Membrane distal domain of MHC.
14. What are memory T cells?
15. Define: Agglutination reaction.
16. Give an example of a radioisotope used to tag antibodies in RIA.
17. State one application of Coomb's test .
18. State true or false: Identification of a specific protein in a complex mixture of proteins can be accomplished by Western blotting.
19. State true or false: Flow cytometer is used to determine the type and number of white blood cells in blood samples.
20. State one application of indirect ELISA.

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- Q 2 A** Discuss the structure and role of lymph node as a lymphoid organ. **08**
- Q 2 B** What is hematopoiesis? Diagrammatically explain the process. **07**
- OR**
- Q 2 C** Describe the alternate pathway for complement activation. **08**
- Q 2 D** What are mononuclear phagocytes? Discuss in brief their structure and function. **07**
- Q 3 A** Describe the structure and functions of TCR complex. **08**
- Q 3 B** Using a neat labelled diagram give the overview of B cell development **07**
- OR**
- Q 3 C** Describe the exogenous pathway for antigen presentation by MHC. **08**
- Q 3 D** Enlist the different types of T cells and describe their functions in brief. **07**
- Q 4 A** Explain the steps involved in ELISPOT assay. **08**
- Q 4 B** Describe two dimensional immunoelectrophoresis. **07**
- OR**
- Q 4 C** What is complement fixation test? Describe the technique. **08**
- Q 4 D** Describe passive agglutination and give its application. **07**
- Q 5** Write short note on **any three** of the following **15**
- a** Primary lymphoid organ.
 - b** NK cells.
 - c** T cell maturation.
 - d** Technique of direct immunofluorescence.
 - e** Technique of ELISA using chemiluminescence.